(Please write your Enrollment Number) Enroll

MID-TERM EXAMINATION
(B.Tech, ECE/ECE-AI/Reappear) (Semest
(March, 2024)

Subject Code: BEC 206	Subject: Electror
Time : 1 ½Hours	
Note: Q. 1 is compulsory.	

Q1			
	(a) Derive spherical coordinates in terms of Cartesian coordi diagram.		
	(b) Discuss Gauss's law.		
	(c) Discuss the conservative nature of an electrostatic field.		
	(d) Explain the difference between line, surface and volume charge density.		
Q2	(Attempt any Two Parts) UNIT-1	(5,5)	
	(a) Derive laplace and poisson's equations. Find $\nabla^2 V$ for the scalar field $V = r^2 (1+\cos\theta \sin\phi)$.		
	(b) Given vector $A = 2a\rho + 3a\phi + 4az$, convert A into Cartesian coordinates at point $(2, \pi/2, -1)$		
	(c) State Stoke's theorem. Evaluate the curl of $A = xy\, a_x + y^2 a_y + xz\, a_z$.		
Q3	(Attempt any Two Parts) UNIT-2	(5,5)	
	(a) Demonstrate the general case of electric fields intensity due to line charge distribution.		
	(b) Prove that tangential component of electric field undergoes no change on the boundary between two different media.		
	(c) Given the potential $V = (10/r^2) in\theta\, cos\phi$. Find the electric flux density D at $(2,\pi/2,0)$		